# Project Proposal: Data Mining I (1DL360)

## Uncovering User Engagement Archetypes and Cross-Platform Integration Patterns on FriendFeed

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## Non-Technical Problem Formulation & Research Questions

Online social networks allow individuals to aggregate content from diverse web services (e.g., microblogs, social bookmarks, and media sharing). However, user activity on such aggregate platforms is highly diverse, ranging from automated bot distribution to organic community conversation. This project addresses two primary questions:

1. **Behavioral Personas:** Can we discover distinct user archetypes (e.g., automated broadcast bots, influential conversationalists, passive lurkers) based on how individuals link external services, publish multimedia, and interact with the community?
2. **Synergy & Discussion Triggers:** Which third-party platforms are naturally adopted together, and what specific content modalities systematically drive active discussions (comment chains) rather than passive acknowledgement (likes)?

## Rationale and Practical Importance

Answering these questions provides actionable value across multiple operational and analytical domains:

- **Automated Moderation & Bot Detection:** Distinguishing genuine community builders from automated feed syndicators (e.g., automated RSS/Twitter cross-posters) improves algorithmic feed curation and reduces platform spam.
- **Content & Marketing Strategy:** Digital creators and community managers gain empirical evidence on which external services complement each other and which posting behaviors spark meaningful group discussions.
- **Platform Design:** Enables system designers to prioritize high-synergy external platform integrations based on real usage.

## Data Specification & Preprocessing Strategy

We use the course-provided **FriendFeed (Version 2)** dataset (Aug–Oct 2010)

- **Data Cleaning & Transformation:** Empty geographic attributes (`GeoX`, `GeoY`) will be dropped, and text encoding/missing profile descriptions in `users.csv` handled.
- **Aggregation (Unit of Analysis):** Because raw data is recorded at the event level, we will aggregate data to the **user level** to construct a multi-dimensional behavioral feature vector (posts count, comments written/received, likes given/received, media frequency, in-degree/out-degree from the follower network).
- **Feature Scaling & Outlier Handling:** Activity attributes follow heavy-tailed distributions. To prevent high-magnitude features (e.g., follower counts) from dominating distance functions, we will evaluate **Standardization ($Z$-score)** and **Robust Scaling** (median/IQR) to safely handle extreme power-users without corrupting proximity metrics.
- **Transaction Formatting:** User-service relationships from `services.csv` will be transformed into binary transaction baskets.

## Proposed Data Mining Methodology

These questions cannot be answered using simple SQL filters or descriptive summary statistics because they require uncovering latent multi-dimensional groupings and non-trivial combinatorial dependencies:

1. **User Archetype Discovery (Clustering & Dimensionality Reduction):**
   - We will apply clustering algorithms (e.g., $K$-Means / Hierarchical Clustering) using Euclidean/Manhattan metrics on scaled user activity vectors to identify organic user personas.
   - To address the curse of dimensionality and visually interpret cluster boundaries, we will apply **Principal Component Analysis (PCA)** to project the feature space into two principal axes of variance.
2. **Cross-Service Patterns (Association Rule Mining / Apriori):**
   - We will execute the **Apriori algorithm** on the service-portfolio transaction data to discover association rules ($X \rightarrow Y$).
   - Candidate pruning (anti-monotonicity of support) will be leveraged to avoid combinatorial explosion ($2^d$ candidate space).
   - Rules will be evaluated using **Support**, **Confidence**, and critically **Lift** ($> 1.0$) to avoid concluding false affinities driven merely by universally popular baseline platforms (e.g., Twitter).

## Expected Results & Validation Plan

- **Cluster Interpretability:** Identification of 3–5 distinct, explainable user personas validated through cluster feature profiles (mean/median comparison per cluster) and 2D PCA scatter plots.
- **Robust Association Rules:** Extraction of non-trivial, high-lift association rules among connected external platforms, with verification that closed and maximal frequent itemsets effectively summarize the itemset space.